Audit Report

Table of Contents

1. Task and dataset understanding

Responsible: Lela; Contributor: Enise

The provided codebase aims to predict hospital patients' mortality using a labeled dataset. This is therefore a supervised binary classification task. The objective is to predict a binary target variable, "hospital_expire_flag," indicating whether a patient survived or died.

The dataset consists of three tables summarized in Table 1. These tables are joined using common identifiers (subject_id for PATIENTS and ADMISSIONS, followed by a merge with ICUSTAYS using hadm_id), producing 136 records and 36 variables for model training. The dataset contains numerical attributes (e.g., subject_id), categorical attributes (e.g., gender), temporal attributes (e.g., dod), and identifier variables (e.g., icustay_id). Identifier variables require careful handling as they may introduce spurious associations if used for prediction.

A preliminary inspection of the dataset identifies several methodological risks such as an imbalanced target variable, missing data, and potential target leakage through post-outcome attributes. Additionally, the dataset contains repeated patients across ICU stays and admissions, which might result in training and test sets being non-independent. These motivate the audit presented in the next sections.

Table 1: Dataset tables and their columns

Table Name	Purpose	List of Columns
Patients	Demographics information	subject_id, gender, dob, dod, dod_hosp, dod_ssn, expire
Admissions	Admission Information	subject_id, hadm_id, admittime, dischtime, deathtime, admission_type, admission_location, discharge_location, insurance, language, religion, marital_status, ethnicity, edregtime, edouttime, diagnosis, hospital_expire_flag, has_chartevents_data
ICU Stays	Information about ICU Stays	subject_id, hadm_id, icustay_id, dbsource, first_careunit, last_careunit, first_wardid, last_wardid, intime, outtime, los

2. Audit Analysis

Responsible: Lela, Enise

This section evaluates the ML pipeline's issues, noting each one's location (original codebase), severity, and impact.

Issue 1: Preprocessing Before Train/Test Split

Location: Lines 120–160

Description: Dropping columns with above-threshold missing value rate, data imputation, outlier clipping, one-hot encoding, and feature scaling are all applied before splitting the dataset into train and test sets.

Justification of Severity: The model has direct access to information unavailable during prediction, invalidating unbiased performance estimates.

Impact: Produced optimistic values for evaluation metrics propagates to GridSearchCV (lines 176–210) since each validation fold is transformed using the statistics from the complete dataset instead of only the respective training fold, making model selection unreliable.

Issue 2: Data Leakage through Post-Outcome Variables

Location: (Table: column names)

- PATIENTS: dod, dod_hosp, dod_ssn
- ADMISSIONS: deathtime, dischtime, discharge_location
- ICUSTAYS: outtime, los
- LOS imputed as numeric feature, clipped and scaled (lines 133–135, lines 138–139, lines 154–155)
- discharge_location one-hot-encoded at lines 142–145

Description: The dataset contains post-outcome variables that are only available after the clinical outcome is known.

Justification of Severity: Inclusion of post-outcome variables invalidates the model by allowing it to directly infer outcomes instead of learning the meaningful mappings between attributes related to mortality risk and outcomes.

Impact: This causes data leakage and produces optimistic generalization metrics. The variable discharge_location == "DEAD/EXPIRED", gives 100% accuracy and a ROC AUC of 1.0 on the full dataset. XGBoost's reported result (0.964 accuracy, 0.994 ROC-AUC) is likely driven mainly by this column, not real patterns.

Issue 3: Repeated Patients and Admission Handling

Location: Lines 19, 22 (join operations)

Description: The split is performed at row level rather than patient level. Some rows in the final table come from the same patient or the same admission and share the same hospital_expire_flag and admission-specific information, only the ICU-specific details differ between them.

Justification of Severity: This issue breaks the assumption of test/train data are independent. Patient overlap indicates test performance might not translate into generalization to new patients, reducing reliability.

Impact: Metrics overestimate generalization since test data isn't independent from training. Affected: 14/136 patients (multiple admissions) and 6 admissions (multiple ICU stays).

Issue 4: Missing Domain Validation

Location: No explicit validation step after EDA

Description: Guardrails such as admittime preceding dischtime, ICU intime preceding outtime, los being nonnegative, plausible age range are not explicitly validated. Aggregate statistics such as describe() summarize central tendencies only and might conceal logical violations along the rows.

Justification of Severity: If invalid records remain undetected, the model may learn relationships that do not reflect real-world cases.

Impact: Undetected inconsistencies introduce incorrect patterns into training data, degrading the model's ability to learn meaningful relationships, reducing reliability.

Issue 5: Outlier Analysis

Location: Lines 138–140

Description: Outlier handling needs to be grounded in methodologies derived from domain or statistical criteria (inspecting boxplots, determining plausible ranges, inspecting IQR thresholds, and visualizing distribution). Clipping arbitrarily at the 99th percentile might remove meaningful observations.

Justification of Severity: Although this issue does not invalidate the pipeline, it may reduce the models' reliability and generalizability.

Impact: Meaningful extreme observations might be removed, which could reduce predictive performance on unseen data.

Issue 6: Missing Value Handling

Location: Lines 33 - 34, 120 - 135.

Description: The pipeline does not check for why values are missing before choosing a method, and the 50% threshold is unexplained. It coincidentally drops deathtime (66% missing, a leakage column), while dod_hosp (35%) and dod_ssn (21%) posing similar risk stay under the threshold and remain.

Justification of Severity: The 50% rule gives a false sense that leakage-related columns have been handled, when it only removes one of them by chance.

Impact: The median and "Unknown" filling could distort the true distribution of the data, and the missing value threshold does not reliably filter out leakage risks.

gIssue 7: Class Imbalance Ignored

Location: Lines 169–210

Description: The train/test split uses stratify=y, which keeps the same class ratio in both sets, but there is no other handling (class weights, resampling). The hospital_expire_flag is imbalanced (69% survived, 31% died). The code checks this distribution, but there is no class weighting and no resampling method (SMOTE or undersampling).

Justification of Severity: It does not invalidate the pipeline, but reduces how reliable and meaningful the results are, especially for the minority class (the death cases).

Impact: The model achieves very low recall (0.333 for Logistic Regression and 0.111 for Random Forest) while precision is 1.0. This indicates that these models predict "survived" almost every time, and only rarely predict "died." In this context, this is a major weakness, because models miss the majority of the patients at risk.

Issue 8: Identifier Variables as Predictive Features

Location: Lines 150–166. Identifier variables (subject_id, hadm_id, icustay_id, etc.) are excluded from scaling but remain in the feature matrix at line 165.

Description: Since identifiers do not represent meaningful clinical characteristics, they do not contribute meaningfully to predictions on new patients and may encourage the model to learn dataset-specific patterns.

Justification of Severity: Identifier variables do not provide predictive information and should not influence the learning. Models might learn spurious associations based on such identifiers, which would lead to poor generalization. In tree-based models (Random Forest), the risk increases as a split threshold might directly be selected on subject_id or hadm_id since they are both numeric columns.

Impact: Retaining identifiers may cause reliance on IDs over clinical data, and would inflate the performance by reducing generalization.

Issue 9: Use of Magic Numbers

Location: Examples include: Line 124 (threshold = 0.5), Line 138 (quantile(0.99)), Line 170 (test_size = 0.2), Line 176 (max_iter=5000, random_state=42), 179 (CV=5)

Description: These hard coded numerical values are embedded directly rather than being explicitly defined as standalone named constants or configuration parameters.

Justification of Severity: It does not affect the ML pipeline but reduces readability, maintainability, and reproducibility. Modifying the values is no longer trivial.

Impact: Future maintenance and experimentation are affected negatively. This might also lead to accidental configuration errors.

3. Severity Classification

Responsible: Lela; Contributor: Enise

Issue	Classification	Justification	Impact
Preprocessing before split	Critical	Test data influences preprocessing and cross-validation,	Performance, Validity, Reliability

Issue	Classification	Justification	Impact
		invalidating evaluation and resulting in optimistic estimates	
Target leakage	Critical	Uses information unavailable at prediction time, allowing direct inference of the target. Such a model is not usable in real-world deployment.	Performance, Validity, Interpretation
Repeated Patients	Major	Violates independence between train and test observations, inflates performance.	Validity, Reliability
Domain validation	Major	Invalid records may bias model learning and make it less reliable	Validity, Reliability, Reliability
Outlier analysis	Major	The threshold lacks statistical justification. This reduces interpretability of preprocessing steps.	Performance, Reliability
Missing values	Major	Thresholds and imputation strategy are insufficiently justified.	Reliability, Interpretation
Class imbalance	Major	Reduces minority-class performance, leading to poor recall for patients at risk	Performance, Reliability, Interpretation
Identifier variables	Major	Encourages learning from meaningless identifiers,	Performance, Validity, Reliability

Issue	Classification	Justification	Impact
		thus reducing model validity and interpretability	
Magic numbers	Minor	Primarily affects maintainability and readability	Interpretation

4. Methodological Critic

Responsible: Enise; Contributor: Lela

The methodology used in the original codebase is not reliable for predicting mortality with high accuracy. According to Issue 3, the row-level split results in some rows sharing the same patient or admission, violating what (Hughes, Data Leakage) refers to as "Failure to Consider Dependent Samples," where GroupKFold is recommended but not used here. The use of a single split for a small test set is unstable: a single split "may favor one model over another" by chance (Hughes, Splitting Data). The code applies 5-fold CV only within hyperparameter search, not to assess final test-performance stability, and since preprocessing was fitted before the split (Issue 1), this same leakage carries into those CV folds too.

The reported metrics are a reasonable selection. Accuracy is biased under class imbalance, and the confusion matrix and ROC-AUC remain more reliable (Hughes, Classification Metrics). Issue 7 highlights this: Random Forest's precision reaches 1.0 while its recall drops to just 0.111, and Logistic Regression shows a milder version of the same problem, with precision 0.750 and recall 0.333. Class weighting and resampling are not used in the pipeline to address the imbalance. Since false negatives (missed deaths) are more dangerous than false positives here, the code never discusses this or justifies the default 0.5 threshold.

Apart from Issue 1, Date columns (admittime, dischtime, etc.) are also one-hot encoded without checking cardinality first: one-hot encoding "increases dimensionality" and should be used carefully (Hughes, Data Preprocessing). ID columns are excluded only from scaling, not from the feature set itself, and remain in the model input (Issue 8). Additionally, missing values are filled with the median or "Unknown" without proper analysis; understanding whether data is missing completely at random or not at random should guide the choice of imputation method (Hughes, Exploratory Data Analysis). The outlier handling also does not follow the IQR or Z-score methods covered in that same lecture (Hughes, Exploratory Data Analysis). StandardScaler is used despite being sensitive to outliers, when RobustScaler would be more appropriate.

Logistic Regression, Random Forest, and XGBoost are reasonable choices for this problem. The real issue is the input. With around 1,217 columns for only 136 patients, all three models risk fitting to noise, and no baseline model exists to establish a chance line. The reported performance is overly optimistic due to leakage, invalid evaluation design, and non-independent samples.

5. Improvement Plan

Responsible: Lela; Contributor: Enise

1. The preprocessing pipeline will be redesigned so that preprocessing operations are fitted only to training data and encapsulated within scikit-learn's Pipeline. This eliminates data leakage.

2. Post-outcome variables (deathtime, dischtime, etc.) and identifiers will be removed to prevent target leakage.

3. Domain-informed validations such as temporal consistency between admission and discharge times will be added. A plausible age range is also recommended, though shifted admission/birth dates in the MIMIC dataset limit reliability and are therefore excluded.

4. Missingness patterns will be inspected before imputation. Informative missingness (e.g., ED timestamps) will be preserved, while imputation statistics will be learned from the training data.

5. IQR-based outlier detection and log1p transformation will be explored and compared with the original 99th percentile clipping to decide the appropriate strategy to handle extreme observations.

6. Categorical features will be one-hot encoded after the train/test split.

7. Class imbalance will be addressed via class weighting, though resampling (SMOTE) will be considered. Splitting will be performed at the patient level for independent evaluation.

8. The existing metrics will be retained, but a threshold selection process will be introduced to ensure alignment with clinical objectives and minimization of false negatives. A majority-class baseline will be added, and repeated cross-validation will replace the single train/test split for a steadier estimate.

9. A config.py file will be introduced to store all configuration variables for readability and maintainability.

6. Results and Reflection

Responsible: Lela, Enise

6.1 Comparison of Original and Revised Pipeline

To achieve more reliable evaluation, we split the data by patient, as opposed to by row, so no patient appears in both train and test. Hyperparameters are tuned via cross-validation on the training set, along with a decision threshold set to catch > 70% of death cases. The test set is held out entirely for final evaluation and never influences model choices. Since the dataset is small (136 rows), we repeat grouped cross-validation multiple times across the full dataset and average results for a more stable performance estimate. However, between our models XGBoost's recall dropped to 0.417 on test, this threshold did not generalize well to the small, different set of test patients.

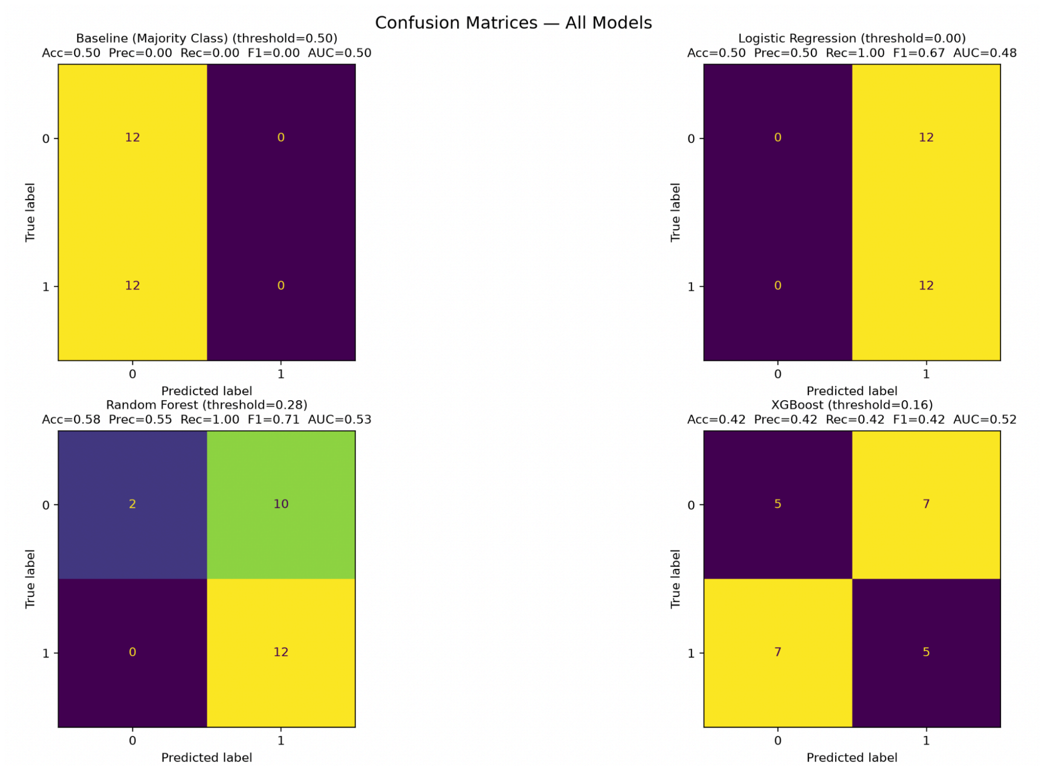

Figure 1: Confusion matrices for the improved pipeline after threshold selection to maximize recall. Complements results in Table 2.

As a result of pipeline imporvements we see that the revised pipeline resulted in lower performance. Given all the issues identified in the section 2, this was an expected result as several features (e.g., *discharge_location* and *deathtime*) directly reveal the outcome. Therefore, removing these variables was the primary contributor to the performance drop while making the model more representative of a real-world deployment. The comparison of model performance is provided in Table 2.

Table 2: Comparison of Model Performance Before and After Pipeline Improvements

Model	Pipeline	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	Original	0.750	0.750	0.333	0.462	0.789
Logistic Regression	Improved	0.500	0.500	1.000	0.667	0.479
Random Forest	Original	0.714	1.000	0.111	0.200	0.924
Random Forest	Improved	0.583	0.545	1.000	0.706	0.531
XGBoost	Original	0.964	1.000	0.889	0.941	0.994
XGBoost	Improved	0.417	0.417	0.417	0.417	0.524

The table below provides an overview of the possible impact of each issue from section 2, ranked by importance.

Table 3: Expected Impact of Pipeline Improvements

Rank	Fix	Expected Impact
1	Target leakage (Issue 2)	Largest impact. Variables such as discharge_location, deathtime, and dod, allowed the model to infer mortality directly. Consequently, performance dropped after removing these variables while producing more realistic evaluations.
2	Preprocessing after train/test split (Issue 1)	Prevented information from the test set influencing imputation, scaling, clipping, and encoding. The effect was smaller than fixing data leakage, but provided a statistically correct pipeline.
3	Patient-level independence (Issue 3)	This fix ensures the same patient does not appear in both train and test, and the repeated cross-validation uses the same grouping. The earlier numbers still had some patient overlap.
4	Class Imbalance (Issue 7)	Class weighting during training and threshold selection during prediction improved sensitivity to minority class. This improvement ensures the model is better aligned with the clinical objective of minimizing missed deaths.
5	Identifier removal (Issue 8)	Prevents models from exploiting identifiers instead of clinically meaningful variables. Particularly beneficial for

Rank	Fix	Expected Impact
		tree-based models, such as Random Forest and for XGBoost, which may create splits on numerical IDs.
6	Domain validation (Issue 4)	No invalid samples were found in the dataset; therefore, addressing this issue did not result in a change to the model's capabilities, but provided important guardrails for possible future samples and improved robustness.
7	Outlier Handling (Issue 5)	Checked during exploration and confirmed the extreme Length of Stay values were real. So instead of clipping them out, they were log-transformed to lower their effect without removing them. Modest effect, mainly on robustness rather than raw accuracy.
8	Missing values (Issue 6)	Missing values in edregtime/edouttime were proved to be informative specifically for ELECTIVE admissions, since those never go through the ED, so a missing flag was added to keep that as an informative signal, allowing the model to retain cliniclally important information.
9	Magic numbers (Issue 9)	No effect on the values of the evaluation metrics. This is more about readability and reproducibility.

6.2 Reflection

The main takeaway from this assignment for us is how mistakes early in the pipeline propagate towards final results, and how one cannot rely on accurate model selection alone to ensure that an ML pipeline is methodologically sound and robust. Additionally, this was a good lesson in how relying only on theoretical knowledge without exploring the dataset and design choices might be misleading. For instance, we initially questioned the outlier clipping (Issue 5) and expected the textbook IQR rule to be the correct fix, but further analysis during Part B showed IQR was actually the less appropriate choice here: it classified 14 length-of-stay values as outliers, most of them plausible, patients with genuinely long ICU stays, compared to only 2 flagged by the 99th percentile, as illustrated in Table 4.

Table 4: Exploratory Comparison of Outlier Detection Methods for Length of Stay (LOS)

Method	Upper Threshold (days)	Observations Flagged
IQR (1.5 × IQR)	9.73	14
99th Percentile	28.96	2
Log-transform	- (no cutoff)	0

Rather than adopting either cutoff, we chose to log-transform the Length of Stay variable instead of clipping it by any threshold, which keeps every observation in the dataset while reducing the influence of the extreme values on the model. Additionally, this assignment highlights how high predictive performance is only one part of the equation and should be taken with a grain of salt: our own revised pipeline's single-split looked reasonable in isolation, but repeated, patient-grouped cross-validation revealed it was close to chance level (0.50), underscoring that a single train/test split is not sufficient evidence of a working model.

Table 5: Single Test-Split vs. Repeated Grouped Cross-Validation (ROC-AUC)

Model	Single Test-Split ROC-AUC	Repeated Grouped CV ROC-AUC (mean ± SD)
Logistic Regression	0.479	0.494 ± 0.161
Random Forest	0.531	0.486 ± 0.093
XGBoost	0.524	0.497 ± 0.099

Overall, the assignment clearly demonstrates that methodological reasoning is just as important as having the appropriate model selected when developing a reliable machine learning systems. Thinking back to the beginning, we would first perform a more thorough exploratory analysis before proposing the improvements to ensure that identified issues are fit to the dataset rather than solely relied on general methodological recommendations. Our experience with outlier analysis was a clear indicator of the importance of such practices.